

Problem 6.20

A certificate of deposit, or CD, is a financial product commonly offered to consumers by banks and credit unions. A CD has a specific, fixed term and, usually, a fixed interest rate. A CD must be held until maturity, at which time the money may be withdrawn together with the accrued interest. However, a type of penalty is imposed for early withdrawal.

A two-year certificate of deposit pays an annual effective rate of 9%. The purchaser is offered two options for prepayment penalties in the event of early withdrawal:

A – a reduction in the rate of interest to 7%

B – loss of three months' interest.

In order to assist the purchaser in deciding which option to select, compute the ratio of the proceeds under Option A to those under Option B if the certificate of deposit is surrendered:

(a) At the end of six months.

(b) At the end of 18 months.

Solution

Let P be the initial deposit, and let t be the time in years at which the CD is surrendered.

Under Option A, the annual effective rate is reduced to 7%, so the proceeds are

$$X = P(1.07)^t.$$

Under Option B, the penalty is the loss of three months' interest. Since three months is 0.25 years, the proceeds are

$$Y = P(1.09)^{t-0.25}.$$

Therefore, the ratio of the proceeds under Option A to those under Option B is

$$\frac{X}{Y} = \frac{(1.07)^t}{(1.09)^{t-0.25}}.$$

For part (a), when $t = 0.5$,

$$\frac{X}{Y} = \frac{(1.07)^{0.5}}{(1.09)^{0.25}} \approx 1.0123.$$

For part (b), when $t = 1.5$,

$$\frac{X}{Y} = \frac{(1.07)^{1.5}}{(1.09)^{1.25}} \approx 0.9937.$$

Since the ratio is greater than 1 in part (a), Option A gives larger proceeds at the end of six months. Since the ratio is less than 1 in part (b), Option B gives larger proceeds at the end of 18 months.